

inlinegraphicx

Includegraphics, with an
automatic inline positioning.

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<https://github.com/cpierquet/latex-packages/tree/main/inlinegraphicx>

Inline insertion,  with automatic adjustments.

Inline insertion,  with automatic adjustments.

Inline insertion,  with *dummybox* if not found.

Inline insertion,  with automatic adjustments but without depth.

Inline  proof with small resizing.

Inline insertion, () with automatic adjustments and choice of optimal height.

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1 Introduction

1.1 Loading, useful packages

In order to load `inlinegraphicx`, simply use:

```
\usepackage{inlinegraphicx}
```

The only loaded package is `graphicx`, the rest of the code is in $\text{\LaTeX}$ 3.

1.2 Usage

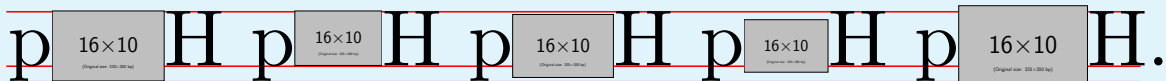
The purpose of this package is to provide command, based on `\includegraphics`, to insert graphic elements *inline*, with automatic positioning and scaling.

Before giving up, the package also tries appending several common extensions to the file name (`.pdf`, `.png`, `.jpg`, `.jpeg`, `.eps`). If the file is still not found, a placeholder image is inserted instead, a small generic icon drawn automatically.

The code determines (total)height and, if necessary, depth of letters in the current font, in order to position the image correctly :

- (total)height is given by (total)height of `qH` in the current font ;
- depth is given by depth of `qH` in the current font.

In order to adjust manually size/positioning, `[keys]` are available.



2 The main macro

2.1 Arguments

The macro for inline insertion is `\inlinegraphics`.

```
\inlinegraphics(*)[scale=...,strut=...]<includegraphics options>{image}
```

The starred version remove depth positioning, wheres normal version include depth.
Available keys are:

- `scale` : re-scaling of compute height (default `1`) ;
- `strut` : characters for height/depth (default `qH`).

2.2 Examples

Inline insertion, `\inlinegraphics{example-image-16x10}` with automatic adjustments.

Inline insertion, `\inlinegraphics[scale=0.75]{example-image-16x10}` with automatic adjustments and scaling.

Inline insertion, `\inlinegraphics*{example-image-16x10}` with automatic adjustments and without depth.

Inline insertion,  with automatic adjustments.

Inline insertion,  with automatic adjustments and scaling.

Inline insertion,  with automatic adjustments and without depth.

```
\scalebox{3}[3]{Inline (\inlinegraphics[strut={()}]{example-image-16x10})
q\inlinegraphics{example-image-16x10}H insertion}.
```

Inline () qH insertion.

```
\sffamily{\huge Inline (\inlinegraphics[strut={()}]{example-image-16x10})
q\inlinegraphics{example-image-16x9}H insertion}.
```

Inline () qH insertion.

```
\ttfamily{\LARGE Inline (\inlinegraphics[strut={(\acute{E})}]{example-image-16x10}\acute{E})
q\inlinegraphics{example-image-16x10}H and q\inlinegraphics{notfoundfile}H insertion}.
```

Inline () $\acute{E}$ qH and qH insertion.

```
\ttfamily{\large Inline insertion,
(\inlinegraphics[strut={()}]<angle=10>{example-image-16x10}) with option given to
includegraphics.}
```

Inline insertion, () with option given to includegraphics.

3 Global configuration

3.1 Purpose

The package needs to know where to look for images in order to run its existence test before calling `\includegraphics`. Three approaches are available, from the most automatic to the most explicit.

3.2 Approach 1 — automatic inheritance from `graphicspath` (default)

By default (`inherit=true`), the package automatically reads the folders declared via `\graphicspath` at `\begin{document}` and uses them for its internal existence test. **No extra macro is needed:** just use `\graphicspath` as usual.

```
\usepackage{inlinegraphicx}
\graphicspath{{images/}{figures/}} % automatically inherited
```

The order of the two lines does not matter — the inheritance is resolved at `\begin{document}`, after all preamble commands have been processed. Note that folder paths must end with a trailing slash (/).

To disable automatic inheritance and manage the search path manually:

```
\usepackage[inherit=false]{inlinegraphicx}
```

3.3 Approach 2 — `inlinegraphicspath` (one macro for both)

`\inlinegraphicspath` uses the same syntax as `\graphicspath` and configures **both** simultaneously: the internal existence test and `\graphicspath` from `graphicx`. Useful when you prefer a single macro to set everything up, or when `inherit=false` is active.

```
\usepackage{inlinegraphicx}
\inlinegraphicspath{{images/}{figures/}}
```

3.4 Approach 3 — `inlinegraphicxsetup` (advanced, decoupled)

`\inlinegraphicxsetup` configures only the internal existence test, without touching `\graphicspath`. Useful when the two need to differ, or to adjust the list of tried extensions.

```
\inlinegraphicxsetup{key1=value1, key2=value2, ...}
```

Available keys:

- `searchpath`: comma-separated list of folders (with trailing /), default empty ;
- `extensions`: comma-separated list of extensions tried when none is given, default `.pdf, .png, .jpg, .jpeg, .eps`.

```
% decouple inlinegraphicx search path from \graphicspath
\usepackage[inherit=false]{inlinegraphicx}
\graphicspath{{figures/}}
\inlinegraphicxsetup{searchpath={images/, figures/}}

% change the tried extensions
\inlinegraphicxsetup{extensions={.pdf, .png}}
```

4 The safeincludegraphics wrapper

4.1 Purpose

`\safeincludegraphics` is a lightweight, fail-safe wrapper around `\includegraphics`. Unlike `\inlinegraphics`, it performs no inline vertical alignment and parses no key of its own: whatever is given as `[options]` is passed verbatim to `\includegraphics`.

```
\safeincludegraphics[options]{file}
```

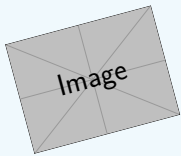
Before inserting the image, the same existence test as `\inlinegraphics` is performed. If the file is still not found, a warning is issued and the `example-image.pdf` placeholder is inserted instead (shipped with the `mwe` package, and available on virtually any L^AT_EX installation, without the need to load `mwe` itself). The placeholder is scaled exactly like a regular image, with the same `[options]`.

4.2 Examples

```
\safeincludegraphics[width=2cm]{example-image-a}
```



```
\safeincludegraphics[width=2cm,angle=15]{thisfiledoesnotexist}
```



5 History

0.20d: Automatic inheritance of graphicspath or macros for an explicit control
0.20c: Macro for params
0.20b: Dummy graphic if file not found
0.20a: Bugfix with gset (tks to D. Bitouze)
0.2.0: LaTeX3 version of code
0.1.1: Pass option to includegraphics within the macro
0.1.0: Initial version

6 The code

```
% Author      : C. Pierquet
% licence     : Released under the LaTeX Project Public License v1.3c or later, see http://www.latex-project.org/lppl.txt

\NeedsTeXFormat{LaTeX2e}
\ProvidesExplPackage{inlinegraphicx}{2026-10-02}{0.20d}{Insert inline graphicx with LaTeXX3}

%====HISTORY
% v 0.20d Macro for path & graphicspath & [inherit=false] option
% v 0.20c Macro for params
% v 0.20b Placeholder if file not found
% v 0.20a Bugfix with gset + l3 improvements
% v 0.2.0 LaTeX3 conversion
% v 0.1.1 Alt options for includegraphics
% v 0.1.0 Initial version

%====PACKAGE
\RequirePackage { graphicx }

%====MESSAGES
\msg_new:nnn { inlinegraphicx } { file-not-found }
{
  Image~'#1'~not~found~;~inserting~a~dummy~image~instead.
}

%====OPTION
\DeclareKeys [ inlinegraphicx ]
{
  inherit .bool_gset:N = \g_inlinegraphicx_inherit_bool ,
  inherit .default:n   = true ,
}
\SetKeys [ inlinegraphicx ] { inherit = true }
\ProcessKeyOptions [ inlinegraphicx ]

%====VARS
\dim_new:N \l_toheight_inlinegraphicx_dim
\dim_new:N \l_depth_inlinegraphicx_dim
\dim_new:N \l_raisebox_inlinegraphicx_dim
\dim_new:N \l_height_inlinegraphicx_dim
\dim_new:N \l_dummyinner_inlinegraphicx_dim
\dim_new:N \l_dummydiag_inlinegraphicx_dim
\dim_new:N \l_dummyht_inlinegraphicx_dim
\dim_new:N \l_dummydp_inlinegraphicx_dim
\dim_new:N \l_dummyraise_inlinegraphicx_dim
\fp_new:N \l_scale_inlinegraphicx_fp
\fp_new:N \l_invscale_inlinegraphicx_fp
\tl_new:N \l_strut_inlinegraphicx_tl
\tl_new:N \l_dummy_inlinegraphicx_tl
\tl_new:N \l_resolved_inlinegraphicx_tl
\clist_new:N \l_extensions_inlinegraphicx_clist
\clist_new:N \l_searchpath_inlinegraphicx_clist
\bool_new:N \l_inlinegraphicx_exists_bool

%====KEYS
\keys_define:nn { inlinegraphics }
{
  scale .fp_set:N      = \l_scale_inlinegraphicx_fp,
  scale .initial:n     = { 1.0 },
  scale .default:n     = { 1.0 },
  strut .tl_set:N      = \l_strut_inlinegraphicx_tl,
  strut .initial:n     = { qH },
  strut .default:n     = { qH },
  dummy .tl_set:N      = \l_dummy_inlinegraphicx_tl,
  dummy .initial:n     = { },
  extensions .clist_set:N = \l_extensions_inlinegraphicx_clist,
  extensions .initial:n = { .pdf, .png, .jpg, .jpeg, .eps },
  searchpath .clist_set:N = \l_searchpath_inlinegraphicx_clist,
  searchpath .initial:n = { },
}

%====HOOK INHERIT
\AtBeginDocument
{
  \bool_if:NT \g_inlinegraphicx_inherit_bool
  {
```

```

\cs_if_exist:NT \Ginput@path
{
  \clist_clear:N \l_searchpath_inlinegraphicx_clist
  \tl_set:Nn \l_tmpa_tl { \Ginput@path }
  \tl_map_function:Nn \l_tmpa_tl
    \inlinegraphicx_inherit_add:n
}
}

%====MACRO CONFIG GLOBAL
\NewDocumentCommand \inlinegraphicxsetup { m }
{ \keys_set:nn { inlinegraphicx } { #1 } }

%====MACRO SEARCHPATH(s)
\NewDocumentCommand \inlinegraphicspath { m }
{
  \graphicspath { #1 }
  \clist_clear:N \l_searchpath_inlinegraphicx_clist
  \tl_map_inline:nn { #1 }
    { \clist_put_right:Nn \l_searchpath_inlinegraphicx_clist { ##1 } }
}

%====INTERNAL FUNCTIONS
\cs_new_protected:Npn \inlinegraphicx_inherit_add:n #1
{ \clist_put_right:Nn \l_searchpath_inlinegraphicx_clist { #1 } }

\cs_new_protected:Npn \inlinegraphicx_check_exist:n #1
{
  \bool_set_false:N \l_inlinegraphicx_exists_bool
  \tl_set:Nn \l_resolved_inlinegraphicx_tl { #1 }
  % 1) raw name
  \file_if_exist:nT { #1 }
  {
    \bool_set_true:N \l_inlinegraphicx_exists_bool
    \tl_set:Nn \l_resolved_inlinegraphicx_tl { #1 }
  }
  % 2) name + ext
  \bool_if:NF \l_inlinegraphicx_exists_bool
  {
    \clist_map_inline:Nn \l_extensions_inlinegraphicx_clist
    {
      \file_if_exist:nT { #1 ##1 }
      {
        \bool_set_true:N \l_inlinegraphicx_exists_bool
        \tl_set:Nn \l_resolved_inlinegraphicx_tl { #1 ##1 }
        \clist_map_break:
      }
    }
  }
  % 3) searchfolder + name (+ ext)
  \bool_if:NF \l_inlinegraphicx_exists_bool
  {
    \clist_map_inline:Nn \l_searchpath_inlinegraphicx_clist
    {
      \file_if_exist:nT { ##1 #1 }
      {
        \bool_set_true:N \l_inlinegraphicx_exists_bool
        \tl_set:Nn \l_resolved_inlinegraphicx_tl { ##1 #1 }
        \clist_map_break:
      }
    }
    \bool_if:NF \l_inlinegraphicx_exists_bool
    {
      \clist_map_inline:Nn \l_extensions_inlinegraphicx_clist
      {
        \file_if_exist:nT { ##1 #1 ###1 }
        {
          \bool_set_true:N \l_inlinegraphicx_exists_bool
          \tl_set:Nn \l_resolved_inlinegraphicx_tl { ##1 #1 ###1 }
          \clist_map_break:
        }
      }
      \bool_if:NT \l_inlinegraphicx_exists_bool { \clist_map_break: }
    }
  }
  \bool_if:NF \l_inlinegraphicx_exists_bool
  { \msg_warning:nnn { inlinegraphicx } { file-not-found } { #1 } }
}

\cs_new_protected:Npn \inlinegraphicx_dummybox:
{
  \dim_set:Nn \l_dummyinner_inlinegraphicx_dim
  { \l_height_inlinegraphicx_dim - 1.2pt }
  \dim_set:Nn \l_dummydiag_inlinegraphicx_dim
  { \fp_eval:n { 1.41421356 * \l_dummyinner_inlinegraphicx_dim } pt }
  \fboxsep 0pt \fboxrule 0.6pt
  \hbox_set:Nn \l_tmpb_box
  {
    \fbox
    {
      \makebox [ \dim_use:N \l_dummyinner_inlinegraphicx_dim ] [ c ]
      {
        \makebox [ 0pt ] [ c ]
        {
          \rotatebox
          [ origin = c ]
          { 45 }
          {

```

```

\rule
{ \dim_use:N \l_dummydiag_inlinegraphicx_dim }
{ 0.5pt }
}
}
\makebox [ 0pt ] [ c ]
{ \rotatebox
[ origin = c ]
{ -45 }
{
\rule
{ \dim_use:N \l_dummydiag_inlinegraphicx_dim }
{ 0.5pt }
}
}
}
}
}
}
% v alignment
\dim_set:Nn \l_dummyht_inlinegraphicx_dim { \box_ht:N \l_tmpb_box }
\dim_set:Nn \l_dummydp_inlinegraphicx_dim { \box_dp:N \l_tmpb_box }
\dim_set:Nn \l_dummyraise_inlinegraphicx_dim
{
0.5 \l_height_inlinegraphicx_dim
-
0.5 \l_dummyht_inlinegraphicx_dim
+
0.5 \l_dummydp_inlinegraphicx_dim
}
\hbox_set:Nn \l_tmpb_box
{
\raisebox
{ \dim_use:N \l_dummyraise_inlinegraphicx_dim }
{ \box_use:N \l_tmpb_box }
}
\box_set_ht:Nn \l_tmpb_box { \l_height_inlinegraphicx_dim }
\box_set_dp:Nn \l_tmpb_box { 0pt }
\box_use:N \l_tmpb_box
}
\cs_new_protected:Npn \inlinegraphicx_insert:nn #1 #2
{
\bool_if:NTF \l_inlinegraphicx_exists_bool
{
\includegraphics
[ height = { \dim_use:N \l_height_inlinegraphicx_dim }, #1 ]
{ \l_resolved_inlinegraphicx_tl }
}
{
\tl_if_empty:NTF \l_dummy_inlinegraphicx_tl
{ \inlinegraphicx_dummybox: }
{
\includegraphics
[ height = { \dim_use:N \l_height_inlinegraphicx_dim }, #1 ]
{ \l_dummy_inlinegraphicx_tl }
}
}
}
}
%====MAIN MACRO
\NewDocumentCommand \inlinegraphics { s O { } D < > { } m }
{
\group_begin:

\keys_set:nn { inlinegraphics }
{
scale = { 1.0 },
strut = { qH },
dummy = { },
}
\keys_set:nn { inlinegraphics } { #2 }

\inlinegraphicx_check_exist:n { #4 }

% ---
\bool_if:NTF #1
{
% --- starred : height = strut height
\hbox_set:Nn \l_tmpa_box { \tl_use:N \l_strut_inlinegraphicx_tl }
\dim_set:Nn \l_toheight_inlinegraphicx_dim { \box_ht:N \l_tmpa_box }
\dim_set:Nn \l_height_inlinegraphicx_dim
{
\fp_eval:n
{
\l_scale_inlinegraphicx_fp * \l_toheight_inlinegraphicx_dim
}
pt
}
\inlinegraphicx_insert:nn { #3 } { #4 }
}
{
% --- non starred : toheight (ht+dp) with v alignment
\hbox_set:Nn \l_tmpa_box { \tl_use:N \l_strut_inlinegraphicx_tl }
\dim_set:Nn \l_toheight_inlinegraphicx_dim
{ \box_dp:N \l_tmpa_box + \box_ht:N \l_tmpa_box }
\dim_set:Nn \l_depth_inlinegraphicx_dim
{ \box_dp:N \l_tmpa_box }
\dim_set:Nn \l_height_inlinegraphicx_dim

```

```

{
  \fp_eval:n
  {
    \l_scale_inlinegraphicx_fp * \l_toheight_inlinegraphicx_dim
  }
  pt
}
\fp_set:Nn \l_invscale_inlinegraphicx_fp
{ 0.5 * ( 1.0 - \l_scale_inlinegraphicx_fp ) }
\dim_set:Nn \l_raisebox_inlinegraphicx_dim
{
  \l_depth_inlinegraphicx_dim
  -
  \fp_eval:n
  {
    \l_invscale_inlinegraphicx_fp * \l_toheight_inlinegraphicx_dim
  }
  pt
}

\raisebox { - \dim_use:N \l_raisebox_inlinegraphicx_dim }
{ \inlinegraphicx_insert:nn { #3 } { #4 } }
}
\group_end:
}

%====SIMPLE WRAPPER
\NewDocumentCommand \safeincludegraphics { 0 { } m }
{
  \group_begin:
  \inlinegraphicx_check_exist:n { #2 }
  \bool_if:NTF \l_inlinegraphicx_exists_bool
  { \includegraphics [ #1 ] { \l_resolved_inlinegraphicx_tl } }
  { \includegraphics [ #1 ] { example-image.pdf } }
  \group_end:
}

\endinput

```